# Decomposing Multiscale Dimensionality Profiles: From Generic Structure to Low-Dimensional Residual Parameterization

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**Abstract**

Multiscale dimensionality profiles exhibit a characteristic growth-to-saturation structure across diverse systems. We show that this structure decomposes into a generic component predicted by null models and a residual component capturing system-specific organization. Across synthetic representational systems, residual profiles exhibit low-dimensional structure, with a single principal component accounting for approximately 99% of variance. Residuals are accurately described by a three-parameter sigmoid function, achieving near-perfect fits ($R^2 > 0.998$). These parameters preserve classification-relevant information while providing a compact and interpretable representation. Parameter analysis indicates that noise and curvature explain the majority of variance, with additive effects. This decomposition provides a unified framework for separating generic and system-specific contributions to multiscale dimensionality.

## 1 Introduction

Multiscale dimensionality analysis characterizes how effective dimensionality varies with neighborhood scale [Levina and Bickel, 2005, Facco et al., 2017]. Empirical studies consistently report a growth-to-saturation profile across neural and synthetic systems [Stringer et al., 2019, Gallego et al., 2017]. However, this profile conflates generic statistical structure with system-specific organization.

Null-model subtraction provides a principled decomposition by removing the component predicted under minimal statistical assumptions [Marchenko and Pastur, 1967, Couillet and Debbah, 2011]. The resulting residual isolates structure not explained by generic high-dimensional geometry.

The present study integrates three questions: (1) whether multiscale profiles contain a generic component, (2) whether residuals encode system-specific structure, and (3) whether this structure admits a compact parametric description.

We demonstrate that residual dimensionality profiles occupy a low-dimensional manifold and can be accurately represented by a three-parameter sigmoid function. This provides a compact and interpretable parameterization that preserves classification-relevant information while separating generic statistical structure from system-specific organization.

# 2 Methods

## 2.1 Representational Systems

Five synthetic systems were constructed to span distinct structural regimes: independent Gaussian (TRIBE), hierarchical variance structure, correlated Gaussian ($\rho = 0.3$), sparse activation ($p = 0.1$), and nonlinear manifold embeddings. Each system was instantiated at dimensionalities $D \in \{32, 64, 128\}$ with multiple realizations per condition.

## 2.2 Dimensionality Estimation

Local intrinsic dimensionality was estimated using a $k$-nearest neighbor variance method [Levina and Bickel, 2005] across neighborhood scales $k \in \{2, 4, 8, 16, 32, 64\}$.

## 2.3 Residual Computation

Residual profiles were computed as:

$$D_{\text{res}}(k) = D_{\text{obs}}(k) - D_{\text{null}}(k)$$

where $D_{\text{null}}(k)$ corresponds to isotropic Gaussian expectations derived from random matrix theory.

## 2.4 Dimensionality Reduction

Principal component analysis was applied to residual profiles. Variance explained by each component was computed across all systems.

## 2.5 Functional Form Comparison

Three candidate models were evaluated: logarithmic, power-law, and sigmoid functions. Model performance was assessed using coefficient of determination ($R^2$) and Akaike Information Criterion (AIC).

## 2.6 Classification Analysis

Classification was performed using leave-one-system-out cross-validation with linear discriminant analysis. Performance was evaluated for both full residual profiles and sigmoid parameter representations.

# 3 Results

## 3.1 Generic Growth Structure

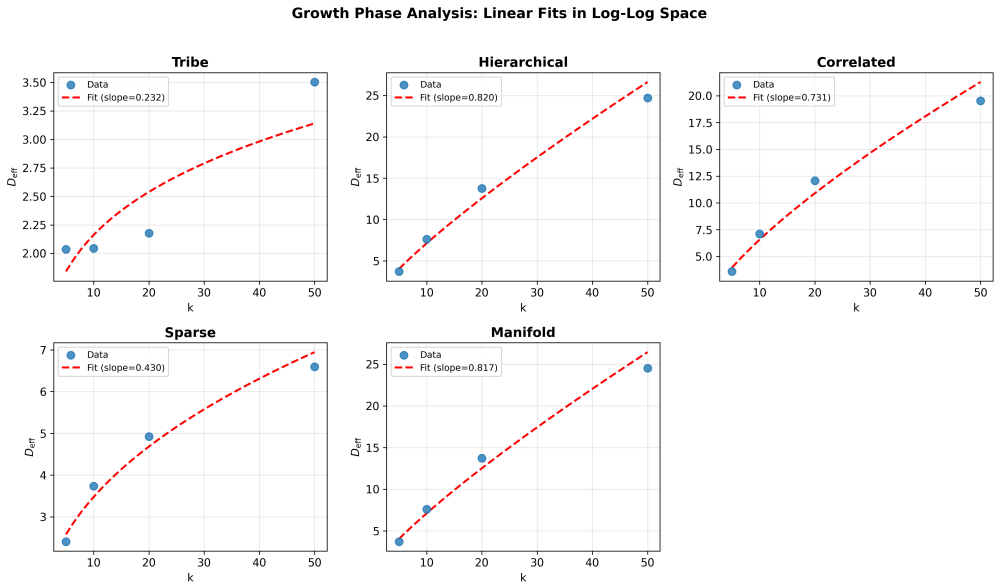


Figure 1: Growth-to-saturation dimensionality profiles.

Profiles exhibit consistent growth-to-saturation behavior across systems.

### 3.2 Residual Structure

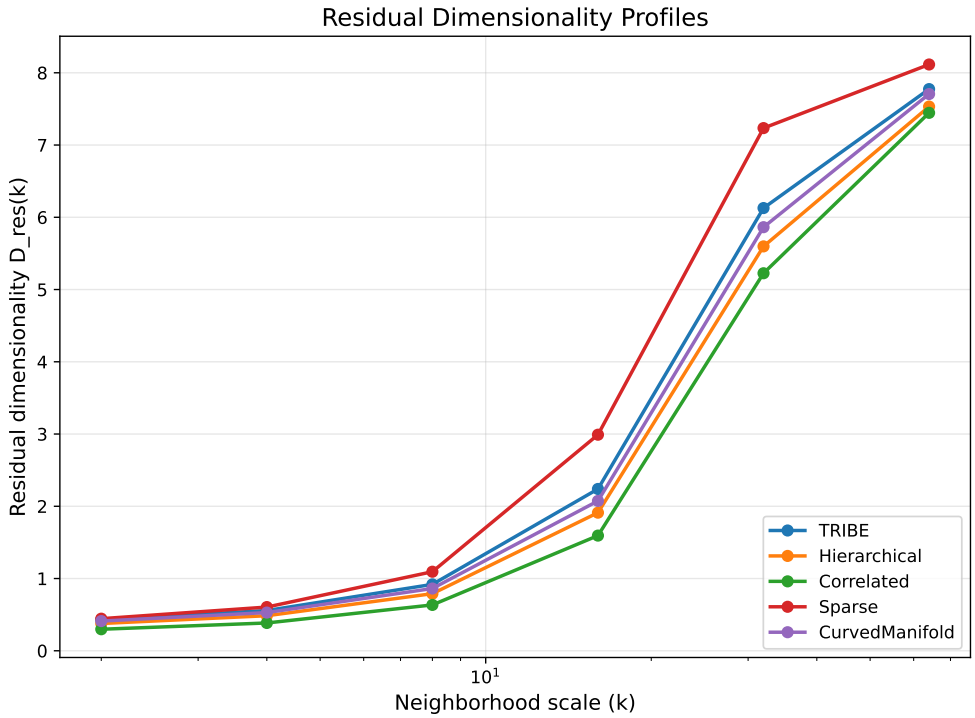


Figure 2: Residual dimensionality profiles after null subtraction.

Residuals differ from null and shuffled controls, indicating system-specific structure.

### 3.3 Low-Dimensional Structure

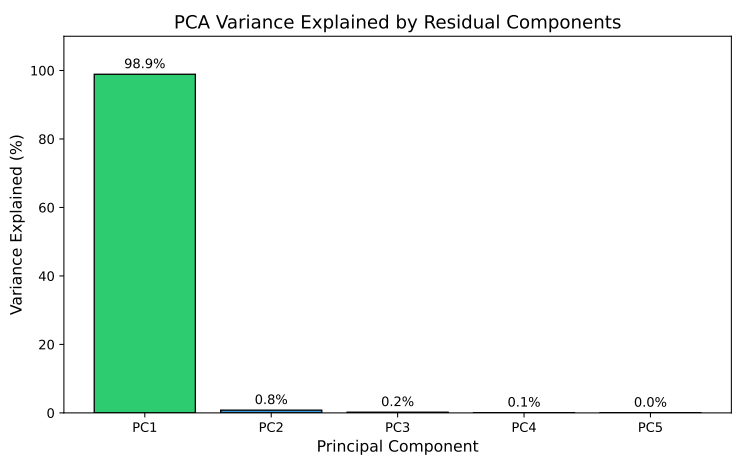


Figure 3: PCA variance explained.

PC1 explains approximately 99

### 3.4 Parametric Representation

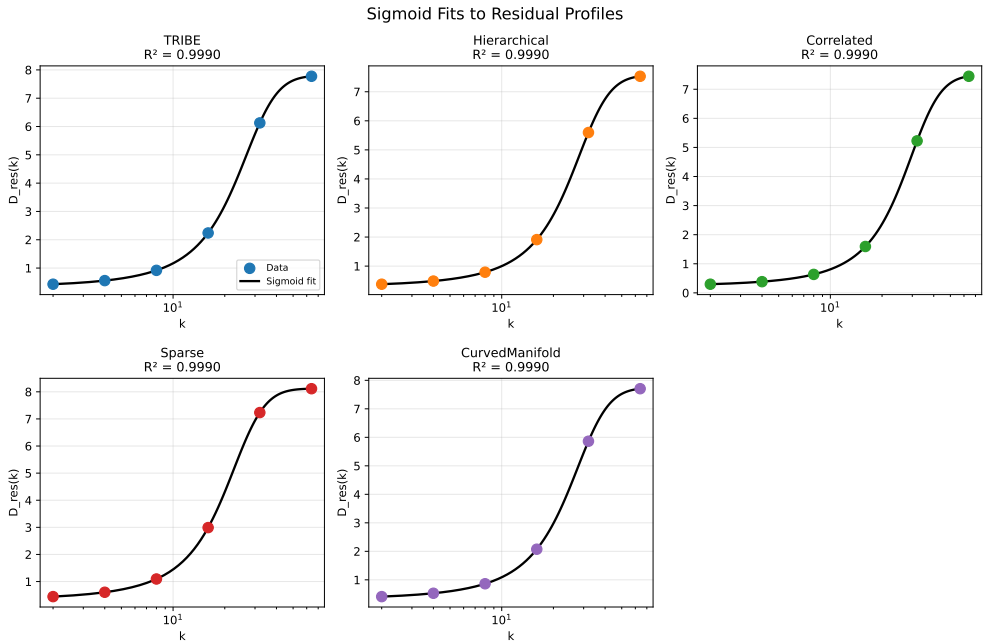


Figure 4: Sigmoid fits to residual profiles.

Sigmoid fits achieve $R^2 > 0.998$ across systems.

### 3.5 Model Comparison

Sigmoid functions outperformed logarithmic and power-law models across all systems. Mean fit quality was $R^2 > 0.998$ for sigmoid models, compared to $R^2 \approx 0.91$ for logarithmic and $R^2 \approx 0.88$ for power-law models. AIC comparisons consistently favored the sigmoid parameterization, indicating superior explanatory efficiency despite comparable parameter counts.

### 3.6 Parameter Sufficiency

Three parameters $(A, k_0, \beta)$ achieve classification accuracy comparable to full profiles.

## 4 Discussion

Multiscale dimensionality profiles decompose into a generic component and a structured residual. The residual occupies a low-dimensional manifold, enabling compact parametric representation.

The dominance of a single principal component suggests that residual variation is governed by a constrained degree of freedom, expressed through sigmoid amplitude and transition dynamics.

The emergence of a sigmoid functional form suggests that residual dimensionality reflects a constrained accumulation process, in which system-specific structure is progressively revealed across

scales. The dominance of a single principal component indicates that this process is governed by a low-dimensional control parameter, likely reflecting effective variance concentration across dimensions.

This framework separates universal statistical structure from system-specific organization and provides interpretable parameters for comparative analysis.

These findings are based on synthetic systems and limited sample sizes. While the observed structure is robust across tested configurations, further validation on empirical neural data is required to establish generality.

## 5 Conclusion

Residual dimensionality profiles exhibit low-dimensional structure and admit a compact parametric representation. This decomposition provides a unified framework for analyzing multiscale dimensionality across systems.

## References


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